

ASSIGNMENT 2

Design and Develop a Web Application using JavaScript

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Batch	A3
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Aim

To design and develop an interactive web application using HTML, CSS and JavaScript, with the use of Array and JSON for storing data and validating user input before saving it.

Objectives

- To develop an interactive web application using JavaScript.
- To use JavaScript Array for storing application data.
- To use JSON for representing the stored data.
- To validate user input before saving the data.
- To display proper error messages for invalid input.
- To handle user actions and events using JavaScript.
- To display and delete registered worker data dynamically.

Theory

1. JavaScript

JavaScript is a scripting language used to make web pages interactive and dynamic. It can respond to user actions such as button clicks, form submission and input events.

In this practical, JavaScript is used to validate the worker registration form, store worker information and display the registered workers on the webpage.

2. JavaScript Array

An Array is used to store multiple values in a single variable. In this application, an Array is used to store the details of registered workers.

The worker details are stored as objects inside the Array.

3. JSON

JSON stands for JavaScript Object Notation. It is a simple format used for representing and exchanging data.

In this practical, the worker data stored in the JavaScript Array is converted into JSON format using `JSON.stringify()`.

4. Form Validation

Form validation is used to check whether the data entered by the user is correct before saving it.

In this application, validation is performed for:

- Name
- Email
- Phone number
- Skill
- Experience
- Location

If the entered data is invalid, an appropriate error message is displayed and the data is not saved.

5. JavaScript Events

Events are actions performed by the user or browser. JavaScript can detect these events and perform a particular operation.

In this practical, the form submission event is used to validate and save worker information. Delete buttons are also used to remove registered workers from the Array.

Application Used

The application developed for this practical is based on Labour Connect: A Digital Platform for Skilled Employment and On-Demand Local Services.

A worker registration application was developed where a skilled worker can enter their personal and professional details.

The application allows the user to:

- Enter worker details.
- Validate the entered information.
- Register a worker if all details are valid.
- Store worker information in a JavaScript Array.
- Convert the Array data into JSON format.
- Display registered workers on the webpage.
- Delete registered worker records.

Software and Hardware Requirements

Software

- Visual Studio Code
- HTML5
- CSS3
- JavaScript
- Web Browser such as Google Chrome or Microsoft Edge
- GitHub

Hardware

- Laptop
- Minimum 4 GB RAM
- Keyboard and Mouse
- Internet connection

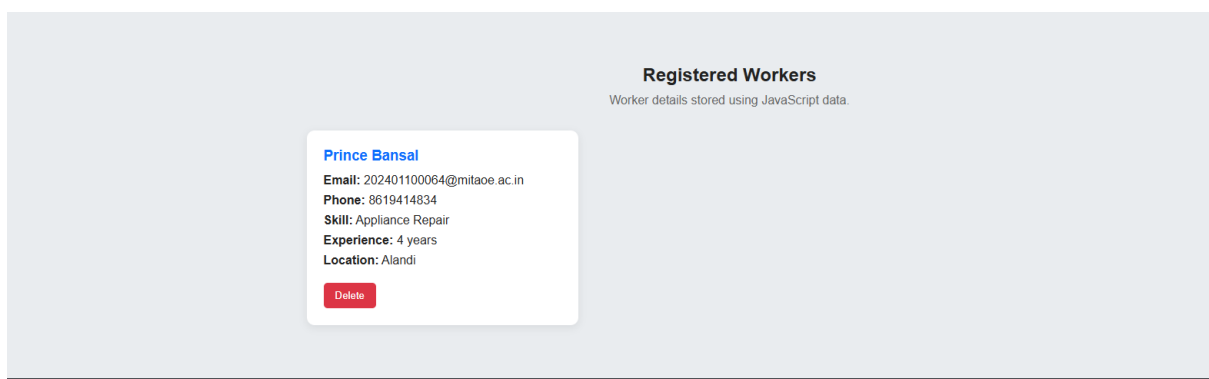
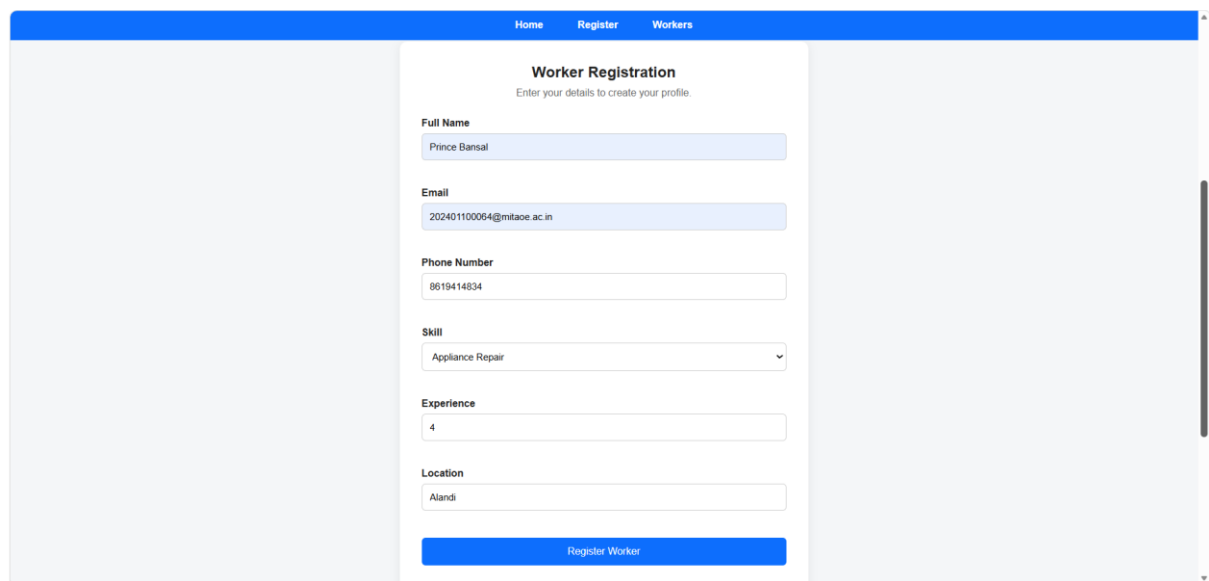
Code

The Practical 2 project files were uploaded to the Web Technology practicals GitHub repository.

GitHub Repository:

[Web-Technology-Practicals/Practical-2 at main · Prince376pb/Web-Technology-Practicals](#)

Output



Worker Registration

Enter your details to create your profile.

Full Name

Prince Bansal

Email

202401100064mitaoe.ac.in

 Please include an '@' in the email address. '202401100064mitaoe.ac.in' is missing an '@'.

Phone Number

08619414834

Phone number must contain 10 digits.

Skill

Select your skill

Please select a skill.

Experience

Experience in years

Enter experience between 0 and 50 years.

Location

Alandi

Register Worker